

Peggy Whitney Altman

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Education

Ph.D. Environmental Science, Colorado School of Mines, Golden, CO

August 2007 – In Progress

Expected Finish: December 2013

Research Focus: *Investigating biologically enhanced dense nonaqueous phase liquid dissolution within fractured bedrock*

M.A.T. Middle Grades Science, Western Carolina University, Cullowhee, NC

August 2005 – July 2007

Research Focus: *Increasing student literacy within the science classroom: How to make science texts more accessible to learners*

B.A. Geology, Washington and Lee University, Lexington, VA

August 2001 – June 2005

Research Focus: *Exploring the interaction of groundwater and bedrock by analyzing chemistry of natural spring water*

Experience

Science and Math Teacher, Vista Academy, Denver Public Schools, Denver, CO

June 2012 – Present

- Develop and implement data driven lesson plans in the areas of AP Environmental Science, Earth Science, Chemistry, Physics, Algebra II, and Financial Algebra.
- Fostered culturally aware classroom environment that builds positive community atmosphere.

Hydrogeology Intern, Itasca Denver, Inc., Lakewood, CO

October 2011 – May 2012

- Data analysis of site data in order to support litigation review of site assessment and remedial actions.
- Assistance of technical report preparation, data entry, and remedial investigations.

Graduate Research Assistant, Colorado School of Mines, Golden, CO

August 2010 – August 2011

- Fellow for the GK-12 partnership between Colorado School of Mines and Aurora Public Schools providing support for local middle school teacher
- Plan and implement innovative, hands-on interdisciplinary lessons for 7th grade literacy, social studies and 6th grade math.

Graduate Research Assistant, Colorado School of Mines, Golden, CO

May 2008 – August 2010

- Investigate relationship between physical properties of fractured systems and the transport, fate, and remediation of chemicals.
- On-going analytical and experimental method development for water chemistry and flow through a network of fractures

Graduate Teaching Assistant, Colorado School of Mines, Golden, CO

August 2007 – May 2008

- Managed teaching laboratory equipment inventory
- Set up lab for Environmental Aqueous Chemistry class

6th Grade Math and Science Teacher, *Asheville Middle School, Asheville, NC*

August 2006 – June 2007

- Implemented differentiated instruction for a variety of learning styles and student disabilities
- Identified and developed effective strategies for “At-Risk” students both within classroom and with team of education professionals

Graduate Research Assistant, *Western Carolina University, Cullowhee, NC*

August 2005 – July 2006

- Assisted Literacy Coach at Asheville Middle School with variety of tasks aimed at increasing student literacy such as on-going student and teacher evaluation, build student library with reading level appropriate options, and led student reading groups to increase student reading efficacy
- Reviewed literature and compiled findings related to effective techniques for improving literacy and worked with teachers to implement best practices in all subject areas

Skills

Analytical: GC-FID, UV-Vis spectroscopy, ICP-AES, GC-MS, pH and ion specific probes

Computing: Transport (CXTFIT, MPNE-1D, HYDRUS 1-D), Chemical Speciation (Visual MINTEQ), Microsoft Office

Other Professional and Leadership Experience

Tutor, *Byrne Urban Scholars, Denver, CO*

January 2010 – June 2012

- Work with Denver area high school students experiencing risk factors for dropping out of school
- Provide homework assistance in both math and science for grades 9-12

Administrative Assistant, *Colorado School of Mines, Denver, CO*

August 2008 – June 2010

- Perform Hydrologic Science and Engineering Division secretarial duties including application processing, filing, and assisting Division director

Senior Guide, *Mountain Adventure Guides, Erwin, TN*

Summers 2003, 2004, 2005

- Planned and guided 12 day backcountry trips for 9-15 year old children
- Organized food, travel, and activities for 10-12 children at a time

Awards and Honors

- Guest Lecturer, Graduate Level Subsurface Contaminant Transport Class, Colorado School of Mines **April 2012**
- Colorado Ground Water Association’s Harlan Erker Memorial Scholarship Recipient **2011 – 2012**
- National Ground Water Association Farvolden Award for Best Student Presentation **May 2011**
- Guest Speaker, Department of Geology, Washington and Lee University **January 2011**
- NC Teach Feature Teacher, Asheville, North Carolina **2006-2007**

Professional Service and Affiliations

Denver Public Schools Next Generation Science Standards Dive Team	January 2014 - Present
Washington and Lee University Alumni Admissions Program Committee	August 2011 - Present
National Ground Water Association	2007 - Present
Geological Society of America	2004 - Present
American Geophysical Union	2004 - Present
National Science Teachers Association	2005 - Present
Environmental Science and Engineering Student Seminar Committee	June 2010 - June 2011
National Ground Water Association 2010 Annual Summit Committee	May 2010 - May 2011
Student Committee for Environmental Science and Engineering	August 2007 - May 2008
Ad Hoc Reviewer for Journal of Contaminant Hydrology	July 2009

Publications

- Altman, P.W.**, K.E. Christensen, C.E. Schaefer, and J.E. McCray. 2014. *Bioenhanced DNAPL dissolution within an experimental 3-D sandstone fracture network*. In preparation for submission to Environmental Science & Technology.
- Altman, P.W.**, K.E. Christensen, J.E. McCray, and C.E. Schaefer. 2014. *Non-reactive transport in a 3-region experimental sandstone fracture network*. In preparation for submission to Environmental Science & Technology.
- Altman, P.W.**, K.E. Christensen, C.E. Schaefer, and J.E. McCray. 2014. *A comparison of DNAPL biological and chemical oxidation mass removal rates in a bench-scale sandstone fracture network*. Planned for preparation for submission to Environmental Science and Technology.
- Christensen, K.E., **P.W. Altman**, C.E. Schaefer, and J.E. McCray. 2014. *DNAPL dissolution kinetics during in-situ chemical oxidation in a three-dimensional experimental fractured sandstone network*. In preparation for submission to Environmental Science and Technology.
- Christensen, K.E., **P.W. Altman**, C.E. Schaefer, and J.E. McCray. 2014. *DNAPL dissolution kinetics in an experimental fracture network*. Submitted for review to Journal of Environmental Engineering.

Presentations and Published Abstracts

- Altman, P.W.**, K.E. Christensen, J.E. McCray, and C.E. Schaefer. 2011. *Biologically Enhanced Tetrachloroethene DNAPL Dissolution in an Experimental Fracture Network*. Presented at National Ground Water Association, Ground Water Summit, Baltimore, Maryland, USA May 1-3, 2011.
- Altman, P.W.** 2011. *Bioremediation of chlorinated solvents in experimental fractured bedrock aquifers*. Invited talk for Geology Department at Washington and Lee University, Lexington, Virginia. January 13, 2011.
- Altman, P.W.**, K.E. Christensen, C.E. Schaefer, and J.E. McCray. 2010. *Biologically enhanced DNAPL dissolution kinetics within bedrock fractures: A bench scale experimental study*. Presented as a poster at National Ground Water Association, Ground Water Summit, Denver, Colorado, USA. April 11-15, 2010.
- Christensen, K.E., **P.W. Altman**, J. King, C. Schaefer, and J. E. McCray. 2010. *Evaluating DNAPL dissolution kinetics during chemical oxidation in an experimental fracture network*. Presented as a poster at National Ground Water Association, Ground Water Summit, Denver, Colorado, USA. April 11-15, 2010.

- Altman, P.W.**, K.E. Christensen, J.E. McCray, and C.E. Schaefer. 2009. *Bioaugmentation of chlorinated solvents in fractured bedrock settings: A bench scale study*. Published abstract 2009 SERDP/ESTCP Partners in Environmental Technology Technical Symposium, Washington, DC, December 1-3, 2009.
- Christensen, K.E., **P.W. Altman**, J. King, C. Schaefer, and J. E. McCray 2009. *Benchscale Fracture Network Experiments: Evaluating DNAPL dissolution rates as a function of DNAPL-water interfacial area and dissolution kinetics during chemical oxidation*. Presented as a poster at the 2009 SERDP/ESTCP Partners in Environmental Technology Technical Symposium, Washington, DC, December 1-3 2009.
- Altman, P.W.**, C.E. Schaefer, K.E. Christensen, and J.E. McCray. 2009. *Investigation of biologically enhanced DNAPL dissolution in fractured rock*. Presented at Geological Society of America, Annual Meeting, Portland, Oregon, USA. October 18-21, 2009.
- Christensen, K.E., **P.W. Altman**, J. King, C. Schaefer, and J. E. McCray. 2009. *Investigating DNAPL dissolution kinetics in an experimental fracture network*. Presented at 2009 Geological Society of America. Portland, OR Oct 18 - 22, 2009.
- Schaefer, C.E., C.W. Condee, **P.W. Altman**, K.E. Christensen, J.D. King, and J.E. McCray. 2009. *DNAPL dissolution during bioaugmentation in fractures*. Presented at the Battelle conference for Remediation of Chlorinated and Recalcitrant Compounds, Baltimore, MD, May, 2009.
- Altman, P.W.**, K.E. Christensen, J.D. King, J.E. McCray, and C.E. Schaefer. 2009. *Microbially enhanced dissolution of DNAPL in a three-dimensional fracture network*. Presented at National Ground Water Association, Ground Water Summit, Tucson, Arizona, USA. April 19-23, 2009.
- Christensen, K.E., **P.W. Altman**, J. King, C. Schaefer, and J. E. McCray, 2009. *Investigation of DNAPL dissolution in a three-dimensional fracture network*. Presented as a poster at 2009 Ground Water Summit. Tucson, AZ April 19 to 24, 2009.

References

John E. McCray, Ph.D.
Colorado School of Mines
Golden, CO
jmccray@mines.edu

Jaime Caldwell
Vista Academy
Denver, CO
Jaime_Caldwell@dpsk12.org

Lisa Greer, Ph.D.
Washington and Lee University
Lexington, VA
GreerL@wlu.edu